INCH-POUND

MIL-C-87074B(NU)

20 March 1991

SUPERSEDING
MIL-C-87074A(NU)
24 October 1984

MILITARY SPECIFICATION

CLOTH, TWILL, ACETATE

This specification is approved for use by the Navy Clothing and Textile Research Facility, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of acetate twill cloth.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, Post Office Box 59, Natick, MA 01760-0001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8305

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2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-P-1133 - Packaging and Packing of Synthetic Fiber Fabrics

STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-2073-1 - DOD Materials Procedures for Development and Application of Packaging Requirements
MIL-STD-2073-2 - Packaging Requirement Codes

* (Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

* 2.1.2 Other Government Documents, Drawings, and Publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

RULES AND REGULATIONS UNDER THE TEXTILE FIBER PRODUCTS IDENTIFICATION ACT

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580.)

2.2 Non-Government publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of documents which are DoD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

TECHNICAL MANUAL OF THE AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS

METHOD No. 135-1987 - Dimensional Changes in Automatic Home Laundering of Durable Press Woven or Knit Fabric

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METHOD No. 8-1988 - Colorfastness to Crocking: Crockmeter Method

* (Application for copies shall be addressed to the AATCC National Headquarters, P.O. Box 12215, Research Triangle Park, NC 27709-9989.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Standard sample. The finished cloth shall match the standard sample for shade and uniformity of shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced. The standard sample is identified under piece number 84288 (see 6.3).

3.2 First article. When specified, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.2).

3.3 Material.

3.3.1 Yarn. The yarn for the warp and filling shall be a continuous multifilament, solution dyed bright acetate. The size of the warp yarn shall be 75 denier and the size of the filling yarn shall be 100 denier when tested as specified in 4.5.

3.3.2 Cloth. The finished cloth shall conform to the physical requirements specified in Table I when tested as specified in 4.5.

Table I - Physical Requirements

Characteristic	
Weight, oz/sq yd (g/m^2)	2.0 min. (67.8)
Yarns per inch (2.5cm)	
Warp	130 min.
Filling	75 min.

3.3.2.1 Weave. The weave for the cloth shall be a 2 up 2 down right hand twill when tested as specified in 4.5.

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* 3.3.2.2 Width. The width of the cloth shall be as specified (see 6.2) and shall be the minimum acceptable width inclusive of selvage when fly shuttle or shuttleless looms, with tuck-in selvage, are used. For all other shuttleless looms the width measurement shall be made between the last warp yarn on each side excluding the protruding fringe.

3.4 Color. The color of the cloth shall be Black 3218 and shall match the standard sample (see 6.3).

* 3.4.1 Matching. The color of the finished dyed cloth shall match the standard sample when viewed under filtered tungsten lamps which approximate artificial daylight, having a correlated color temperature of 7500 (± 200) K, with illumination of 100 (± 20) foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300 (± 200) K.

* 3.4.2 Colorfastness. The finished cloth shall show fastness to laundering, wet-dry cleaning (two methods), dry cleaning, dry crocking, perspiration and light (80 SFH) equal to or better than the standard sample. As a limit of acceptability or when no standard sample is available, the finished cloth shall show a minimum of "good" fastness to laundering, wet-dry cleaning (two methods), dry cleaning, perspiration and light (80 SFH) and shall show a crocking rating of not lower than 3.0 dry when tested as specified in 4.5.

* 3.5 Dimensional stability. The cloth shall not shrink or elongate more than 4.0 percent in the warp and filling when tested as specified in 4.5.

3.6 Non-fibrous material. The starch and protein content inclu trichlorotrifluoroethane soluble and water-soluble material of the finis cloth shall not exceed 8.0 percent when tested as specified in 4.5.

3.7 Acidity (pH). The pH value of the water extract of the finished cloth shall be no less than 5.0 nor more than 8.5 when tested as specified in 4.5.

* 3.8 Length and put-up Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths each not less than 50 yards and shall be put-up in rolls, full width, as specified in PPP-P-1133.

3.9 Fiber identification. Each roll of cloth shall be labeled or ticketed for fiber content in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act.

3.10 Workmanship. The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards when calculated as specified in section 4 shall not exceed the established maximum point value.

4 QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the government to acceptance of defective material.

4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. When required, the first article of finished cloth submitted in accordance with 3.2, shall be visually inspected for color, finish, and appearance and shall be tested for chemical and physical properties in accordance with the applicable methods specified in 4.5.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents.

4.4.2 Examination of the end item. Examination of the end item shall be in accordance with 4.4.2.1 through 4.4.2.5

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* 4.4.2.1 Yard-by-yard examination. Each roll in the sample shall be examined on one side only. When the total yardage in the roll does not exceed 100 yards (91.4m), the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards (91.4m), only 100 yards (91.4m) shall be examined. All defects as defined in Section III of FED-STD-4, which are clearly noticeable at normal inspection distance (3 feet) (.91m) shall be scored and assigned demerit points as listed in 4.4.2.2, except that only those slubs and knots which exceed the limits shown on Sears Fabric Defect Scale (see 6.4), "C" or "4" as applicable for slubs and "B" for knots, shall be scored. No linear yard (increments of 1 yard (.91m) on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than 4 points. The sample size shall be in accordance with the list below. The lot shall be unacceptable if the points per 100 square yards ($83.6m^2$) of the total yardage examined exceeds 25.0 points. The lot shall be unacceptable if the points per 100 square yards ($83.6m^2$) of two or more individual rolls exceed 38.0 points. If one roll exceeds 38.0 points per 100 square yards ($83.6m^2$), a second sample of 20 rolls shall be examined only for individual roll quality examination. The lot shall be unacceptable if one or more rolls in the second sample exceeds 38.0 points per 100 square yards ($83.6m^2$). Point computation for lot quality and individual roll quality shall be shown below.

Lot size (yards)	Sample size (rolls)
3200 or less	8
3201 to 10,000	13
10,001 and over	20

$$\frac{\text{Total points scored in sample size} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{Total yards inspected}} = \text{Points per 100 square yards}$$

4.4.2.2 Demerit points. Demerit points shall be assigned as follows:

For defects up to and including 3 inches (7.6cm) in any dimension - one point
For defects exceeding 3 inches (7.6cm) but not exceeding 6 inches (15.2cm) in any dimension - two points
For defects exceeding 6 inches (15.2cm) but not exceeding 9 inches (22.90cm) in any dimension - three points
For defects exceeding 9 inches (15.2cm) in any dimension - four points

The following defects, when present, shall be scored 4 points for each yard in which they occur:

Baggy, ridgy or wavy cloth
Width less than specified
Characteristics of finish, hand, or shade not equal to the standard sample

4.4.2.3 Examination for length.

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4.4.2.3.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified, or more than two yards (1.8m) less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.4.2.3.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets.

4.4.2.4 Examination for shade.

4.4.2.4.1 Individual rolls. A sample from each roll in the lot shall be examined visually for shade (see 3.4). A roll shall be unacceptable if it fails to meet the requirements for shade, finish or appearance. The sample shall be a 4-inch by 20-inch swatch of cloth drawn from each roll in the lot.

4.4.2.4.2 Examination for shade variation. During the yard-by-yard examination, each roll in the sample shall be examined for shade variation. Any roll in the sample exhibiting uneven shade, shaded variation side to side, side to center or end to end shall be cause for rejection of the entire lot represented by the sample.

4.4.2.5 Examination for noncompliance with the Textile Fiber Products Identification Act. During the yard-by-yard examination, each roll in the sample shall be examined for this defect. The lot shall be unacceptable if two or more rolls in the sample are not labeled or ticketed in accordance with the Textile Fiber Products Identification Act.

4.4.3 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-1133 to determine that packaging, packing and marking comply with Section 5 requirements of this specification.

4.5 Testing of the end item. The methods of testing specified in FED-STD-191 wherever applicable, and as listed in Table II shall be followed. The physical and chemical values specified in Section 3, except where otherwise specified, apply to the average of the determinations made on the sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values utilized in expressing the final result. The sample shall be expressed in units of one yard each. The sample unit shall be 4 yards, full width of the finished cloth. The lot shall be unacceptable if one or more units fail to meet any test requirements specified. The sample size shall be in accordance with the following:

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Lot size (yards)	Sample size (units)
800 or less	2
801 thru 22,000	3
22,001 and over	5

*Table II - Test methods

Characteristics	Requirement paragraph	Test method
Yarns	3.3.1	1/
Weight	3.3.2	5041
Yarns per inch		
Warp	3.3.2	5050
Filling	3.3.2	5050
Weave	3.3.2.1	Visual 2/
Colorfastness to:		
Laundering	3.4.2	5614
Wet-Dry Cleaning	3.4.2	5622
Wet-Dry Cleaning	3.4.2	5621
Dry Cleaning	3.4.2	5620
Crocking	3.4.2	AATCC TM 8-1988
Perspiration	3.4.2	5680
Light(80 SFH)	3.4.2	5660 3/
Dimensional stability	3.5	AATCC TM 135-1987 5/
Nonfibrous materials	3.6	2611 4/
pH	3.7	2811

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirements.

2/ One determination shall be made from each sample unit and the results shall be reported as "Pass" or "Fail".

3/ The calibration of the fadeometer and the definition of AATCC Fading Units shall be according to AATCC Test Method 16-1982. Using the formula below, 16 to 22 AFU's are acceptable as 20 SFH's.

$$\text{AATCC Fading Units} = \frac{(\Delta E^*)\ (20)}{1.7}$$

Where (ΔE^*) is the color difference in CIELAB* units of color difference.

4/ Nonfibrous shall be determined in accordance with the method prescribed except that trichlorotrifluoroethane shall be used in lieu of chloroform.

5/ Machine cycle-permanent press, wash temperature IV (120°F ± 5°), drying procedure (b).-tumble dry. One complete cycle only

5. PACKAGING

* 5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

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* 5.1.1 Levels A and Commercial. The cloth, put up as specified, shall be in accordance with the applicable requirements of PPP-P-1133.

* 5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

* 5.2.1 Levels A, B, and Commercial. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1133.

* 5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1133.

5.4 MIL-STD-2073. When specified (see 6.2), Preservation and Packing shall be in accordance with MIL-STD-2073-1 and MIL-STD-2073-2.

* 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cloth covered by this specification is intended for use in the manufacture of neckerchiefs for Navy personnel.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. When first article is required (see 3.2). The item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, quantity, and testing and approval of the first article.
c. Width required (see 3.3.2.2).
d. Length required if other than specified (see 3.8).
e. Levels of preservation and packing required (see 5.1 and 5.2).
f. When all packaging data is to be found in MIL-STD-2073-1 and MIL-STD-2073-2 (see 5.4)

6.3 Standard sample. For access to the standard sample (see 3.1), address the procuring activity issuing the invitation for bids.

* 6.4 Fabric defect scales Sears Fabric Defect Scales are available from Sears, Roebuck and Company, Department 817, (Attn: BSC 23-29), Sears Tower, Chicago, IL 60684 (see 4.4.2.1).

* 6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

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6.7 Changes from previous issue. The margin of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

* 6.8 Subject term (key word) listing.

Fabric, black neckerchief
Neckerchief, fabric

Custodian:
Navy - NU

Preparing activity:
Navy - NU

Review activity:
DLA - CT

Project 8305-N348

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-87074B(NU)	CLOTH, TWILL, ACETATE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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